Deep Blue Sea: Practice Test 1

Time: 50 minutes. Work with your partner at 14 stations, about 3 1/2 minutes each. When the coach calls 'Rotate!', move to the next station; you may not go back. Bring only pencils: no notes, flashcards, or books, just like the tournament.

Partner names: _____ and _____ Score: _____

Station 1: Dive to the bottom

A submarine dives straight down from the surface to the bottom of the deepest trench, about 11,000 meters down.

1. Write the 5 depth zones in the order the submarine passes through them. Use the science names.

2. The submarine is at 2,500 meters. Which zone is it in?

◯ A Mesopelagic (twilight) ◯ B Bathypelagic (midnight) ◯ C Abyssopelagic (abyss) ◯ D Hadalpelagic (trenches)

3. The submarine is at 150 meters. Which zone is it in? Give the science name and the nickname.

4. At about what depth does the hadalpelagic zone begin?

◯ A 200 m ◯ B 1,000 m ◯ C 4,000 m ◯ D 6,000 m

Station 2: Zone readings

A robot took 3 readings. Reading X: 25 C (77 F), bright light, pressure about 5 times the surface. Reading Y: 4 C (39 F), no sunlight, pressure about 250 times the surface. Reading Z: 1 C (34 F), no sunlight, pressure about 1,000 times the surface.

5. Which reading was taken the deepest? Give 2 clues from the data.

6. Which reading could have phytoplankton (plant-like plankton) growing nearby? Why?

7. As you go deeper, which 2 things go down and which 1 thing goes up?

8. Why is food so hard to find at Reading Y and Reading Z?

Station 3: Fast swimmer

This big, fast fish hunts near the surface of the open ocean.



9. Which zone does this fish mostly live in?

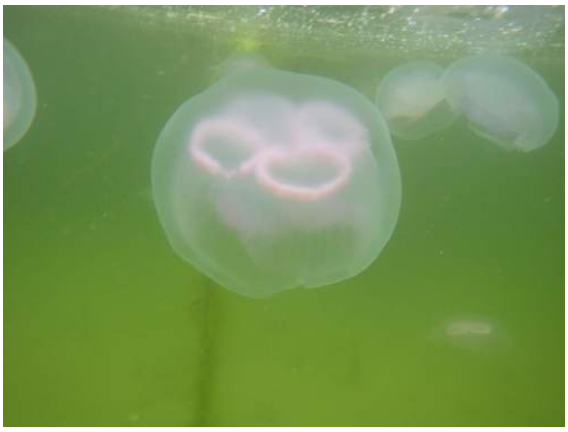
- ◯ A Epipelagic (sunlight)
- ◯ B Mesopelagic (twilight)
- ◯ C Bathypelagic (midnight)
- ◯ D Abyssopelagic (abyss)

10. Its back is dark and its belly is light. What is this adaptation called?

11. How does that coloring hide it from an animal looking down from above AND from an animal looking up from below?

Station 4: Drifters

This see-through animal drifts with the ocean currents.



12. Living things that drift with the currents instead of swimming against them are called _____.

13. Put this food chain in order, starting with the living thing that makes its own food: tuna, zooplankton, small fish, phytoplankton.

14. Is a jellyfish a fish? Explain.

Station 5: Twilight hunter

This fish dives into the dim, cold twilight zone during the day to hunt squid.



15. The twilight zone goes from _____ meters to _____ meters.

16. Can plants or algae grow in the twilight zone? Why or why not?

17. This swordfish has a special trick for seeing and thinking in cold water. What is it?

Station 6: Seeing red

This crab is bright red. It lives on the sea floor at about 600 meters deep.



18. Which color of sunlight does sea water soak up first, near the top?

◯ A Blue ◯ B Green ◯ C Red ◯ D Violet

19. Why is bright red a good hiding color at 600 meters?

20. The only sunlight left at 600 meters is very dim. What color is it?

Station 7: Belly lights

This small, silver fish has rows of light spots on its belly. Its big eyes point up.



21. Making light with chemicals inside the body is called _____.

22. A hungry fish swims below this fish and looks up. How do the belly lights keep this fish safe? Name the adaptation.

23. Why do its big eyes point up?

Station 8: Big mouth

This black fish lives where there is no sunlight at all.



24. Which zone does this fish live in?

- ◯ A Epipelagic (sunlight)
- ◯ B Mesopelagic (twilight)
- ◯ C Bathypelagic (midnight)
- ◯ D Abyssopelagic (abyss)

25. Name one body feature you can see that helps it eat where food is rare. How does it help?

26. There is no sunlight in this zone. What is the only light there?

Station 9: Why glow?

For each animal, choose what its light is used for.

27. An anglerfish waves a glowing ball in front of its mouth.

○ A Lure prey ○ B Hide ○ C Startle a predator ○ D Find a mate

28. A squid is grabbed. It squirts a cloud of glowing ink and zooms away.

○ A Lure prey ○ B Hide ○ C Startle a predator ○ D Find a mate

29. A lanternfish flashes a light pattern that only its own kind uses.

○ A Lure prey ○ B Hide ○ C Startle a predator ○ D Find a mate

30. A shrimp's belly lights match the dim light above it.

○ A Lure prey ○ B Hide ○ C Startle a predator ○ D Find a mate

Station 10: Walking on the mud

A deep-sea robot filmed this sea cucumber walking on the soft, muddy sea floor 5,000 meters down.



31. Which zone is it in? Give the science name.

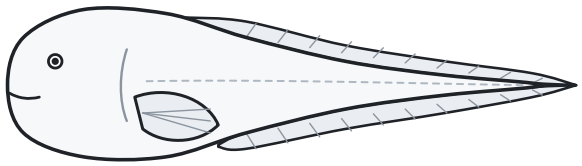
32. No plants grow here. What food drifts down to the mud for it to eat?

33. Many abyss animals move slowly. How does being slow help them survive?

34. Name 2 other animals that live in the abyss.

Station 11: The trenches

This pale fish lives about 8,000 meters down in the Mariana Trench.



35. Which zone does it live in? Give the science name.

36. The water pressure here is huge. Which feature helps this fish survive it?

◯ A A thick, hard shell ◯ B A big air bag (swim bladder) ◯ C A soft, watery body with no air spaces ◯ D Long, sharp teeth

37. Scientists send a foam cup down to the trench. It comes back up much smaller. Why?

38. Tube worms live by hot vents on the deep sea floor. Where does their food come from?

Station 12: Night shift

Scientists used sonar to follow a big group of lanternfish and krill. Depth of the group: 12 noon, 500 m. 6 pm, 300 m. Midnight, 50 m. 6 am, 250 m. 9 am, 450 m.

39. At what time was the group closest to the surface?

40. What is this up-and-down movement every day called?

41. Why does the group swim up at night?

42. Why does the group go back down to about 500 meters in the day?

Station 13: From the shore out

Walk from the beach into the sea: first the shore where the tide comes and goes, then shallow water over the continental shelf, then the deep open ocean past the shelf.

43. Name the 3 zones in order from the shore out.

44. Name 2 problems intertidal animals face.

45. How does a mussel keep from being washed off a rock by waves?

46. Most of the world's fishing happens in the neritic zone. Why is it so full of life?

Station 14: Mystery animals

Read the clues. Write the zone for each animal and the clue that told you.

47. A shark with a dark blue back and a white belly swims near the surface.

48. A black fish with needle-like teeth and a glowing chin whisker lives at 2,000 m.

49. A small snail clamps its shell tight to a rock until the water comes back.

50. A pink shrimp-like animal scavenges dead fish at 9,000 m.

25	Its huge mouth (and stretchy stomach) lets it swallow any food that comes by, even big prey, so it does not miss a meal.
26	Light made by animals (bioluminescence).
Station 9: Why glow?	
27	A Lure prey.
28	C Startle (confuse) a predator.
29	D Find a mate (signal its own kind).
30	B Hide (counter-illumination).
Station 10: Walking on the mud	
31	Abyssopelagic (the abyss), 4,000 to 6,000 m.
32	Marine snow: bits of dead plankton, poop, and dead animals sinking from above.
33	Moving slowly uses very little energy, so they need less food in a cold place where food is rare.
34	Any 2: brittle stars, tripod fish, sea stars, shrimp.
Station 11: The trenches	
35	Hadalpelagic (the trenches).
36	C A soft, watery, jelly-like body with no air spaces, so the pressure is the same inside and out. (It is a snailfish.)
37	The huge water pressure squeezed the air out of the tiny spaces in the foam, so the cup shrank.
38	From bacteria living inside them. The bacteria make food from chemicals in the vent water (chemosynthesis), not from sunlight.
Station 12: Night shift	
39	Midnight (50 m).
40	Daily (diel) vertical migration.
41	To eat the plankton near the surface while the dark hides them from predators.
42	Daylight would let predators that hunt by sight see them. The dim twilight zone keeps them hidden.
Station 13: From the shore out	
43	Intertidal, neritic, oceanic.
44	Any 2: drying out at low tide, crashing waves, hot sun, changing temperature or saltiness, birds and other predators at low tide.
45	It holds on with strong, tough threads (byssal threads).
46	It is shallow and sunny, and nutrients wash in from land, so lots of plankton grow and feed many fish.
Station 14: Mystery animals	
47	Sunlight (epipelagic). Clue: countershading, near the surface.
48	Midnight (bathypelagic). Clue: black body, glowing lure, sharp teeth, 2,000 m.
49	Intertidal. Clue: holds on to a rock and closes up when the tide goes out.
50	Trenches (hadalpelagic). Clue: deeper than 6,000 m; it is like an amphipod.